Functions & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating functions

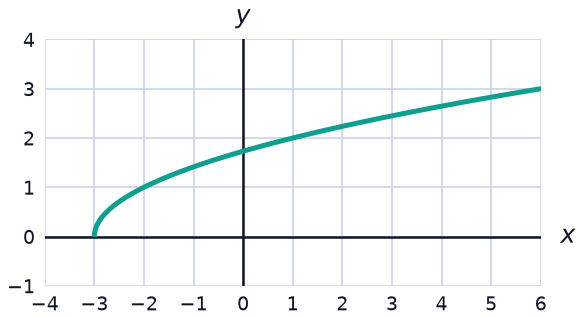
Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:

a $f(0)$

b $f(2)$

2 [Foundational] • Target: Domain and range

The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.



3 [Foundational] • Target: Domain and range

Find the domain of $f(x) = \frac{1}{x - 5}$.

4 [Foundational] • Target: Composition of functions

If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.

5 [Foundational] • Target: Composition of functions

If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

6 [Foundational] • Target: Inverse functions

Find the inverse of $f(x) = 4x - 8$.

7 [Foundational] • Target: Inverse functions

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x - 1}{5}$ are inverses of each other by computing $f(g(x))$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Transformations of graphs

The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f.

9 [Intermediate] • Target: Transformations of graphs

If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g.

10 [Intermediate] • Target: Piecewise functions

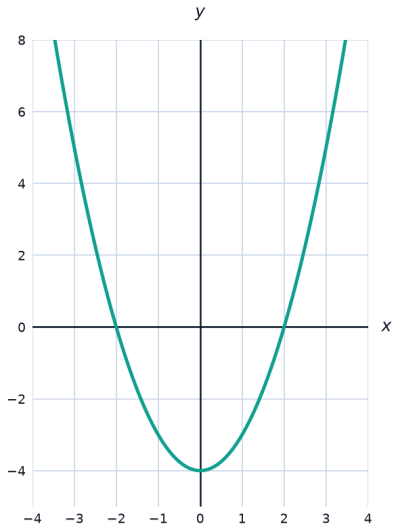
A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

11 [Intermediate] • Target: Piecewise functions

A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.

12 [Intermediate] • Target: Reading function graphs

The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x-intercepts and y-intercept, then find them exactly.



13 [Intermediate] • Target: Even, odd, and other function properties

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.

14 [Intermediate] • Target: Even, odd, and other function properties

Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A well-designed word problem: modeling a car rental cost • Cross-Lesson: 4.01 Ratios, Rates & Data

This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

16 [Challenging] • Target: Reading function graphs • Cross-Lesson: 5.01 Statistics & Probability

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x - 12}}{x^2 - 25}$. Express your answer in interval notation.

17 [Challenging] • Target: Even, odd, and other function properties • Cross-Lesson: 6.01 Geometry

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

18 [Challenging] • Target: A well-designed word problem: modeling a car rental cost • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Let $f(x) = \frac{2x + 1}{x - 3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

19 [Challenging] • Target: Evaluating functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Let $f(x) = \frac{2x - 7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

20 [Challenging] • Target: Domain and range • Cross-Lesson: 1.01 Heart of Algebra

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x - 5) + 2$, determine the corresponding points on the graph of $g(x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating functions

What is the domain of the real-valued function $f(x) = \sqrt{2x - 14}$?

a. $x \geq -7$

b. $x \leq 7$

c. $x > 7$

d. $x \geq 7$

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22 [Foundational] • Target: Domain and range

If $f(x) = \frac{2x+3}{x-4}$ for $xeq4$, what is the inverse function $f^{-1}(x)$?

a. $\frac{4x+3}{x-2}$

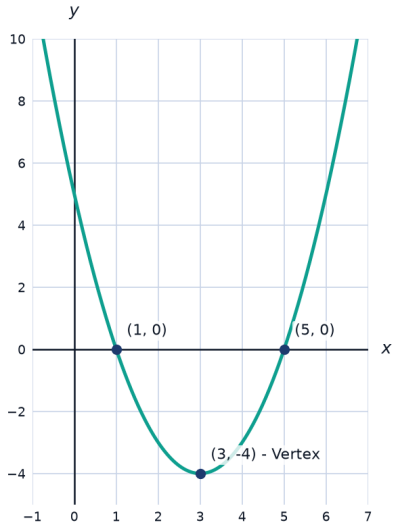
b. $\frac{x-4}{2x+3}$

c. $\frac{4x-3}{x+2}$

d. $\frac{2x-3}{x+4}$

23 [Foundational] • Target: Composition of functions

The graph of the function $y = ax^2 + bx + c$ is shown below. The function passes through the points $(1, 0)$ and $(5, 0)$, and its vertex has a y-coordinate of -4. Determine the value of b.



a. -6

b. 6

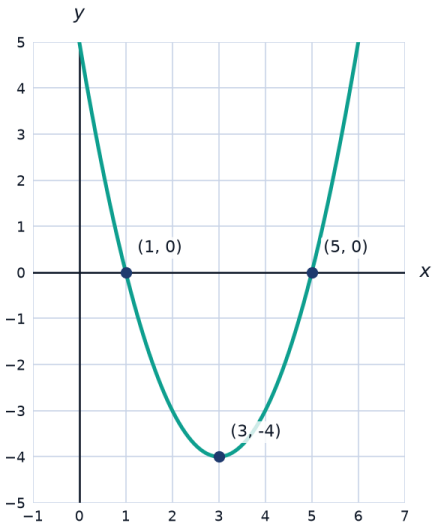
c. -4

d. 4

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24 [Foundational] • Target: Inverse functions

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?



a. 1

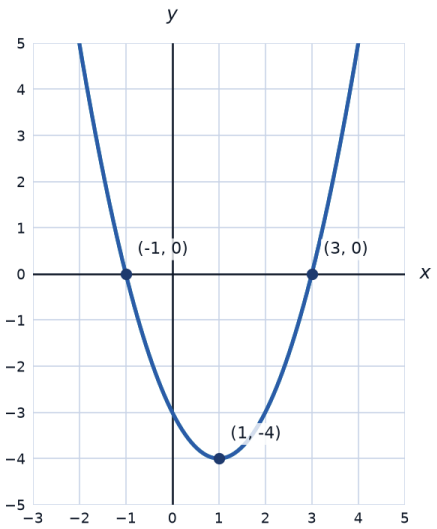
b. -1

c. 2

d. -2

25 [Foundational] • Target: Transformations of graphs

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-1, 0)$ and $(3, 0)$. The vertex of the parabola is at $(1, -4)$. Determine the value of a.



a. -1

b. 1

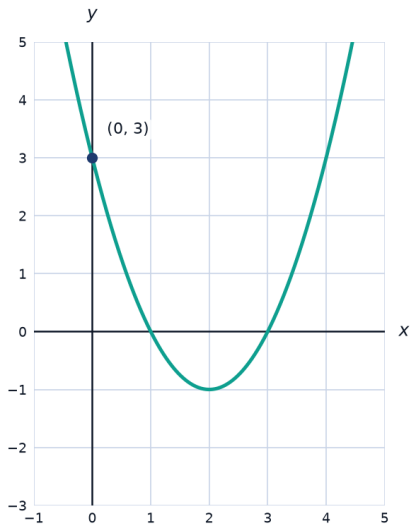
c. 2

d. -2

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26 [Foundational] • Target: Piecewise functions

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(0)$?



a. -3

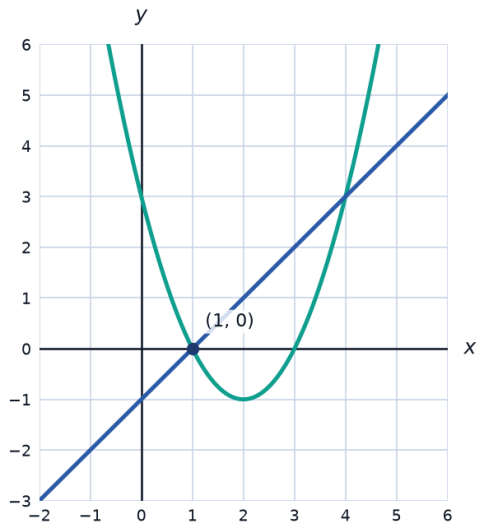
b. -1

c. 1

d. 3

27 [Foundational] • Target: Reading function graphs

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(1)$?



a. 0

b. -1

c. 2

d. 3

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28 [Foundational] • Target: Even, odd, and other function properties

Consider the function $f(x)$ defined by $f(x) = |x - 2| + |x + 1|$. Find the minimum value of $f(x)$ and the interval over which this minimum is attained.

a. Minimum value 3 on the interval $[-1, 2]$

b. Minimum value 2 on the interval $[-1, 2]$

c. Minimum value 3 on the interval $(-\infty, -1]$

d. Minimum value 2 on the interval $[2, \infty)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: modeling a car rental cost

The graph of function g is obtained by shifting the graph of f 6 units to the right and 6 units down. Which of the following defines $g(x)$?

a. $g(x) = f(x - 6) - 6$

b. $g(x) = f(x - 6) + 6$

c. $g(x) = f(x + 6) - 6$

d. $g(x) = f(x + 6) + 6$

30 [Intermediate] • Target: Evaluating functions

The graph of function g is obtained by shifting the graph of f 2 units to the left and 5 units down. Which of the following defines $g(x)$?

a. $g(x) = f(x + 2) - 5$

b. $g(x) = f(x - 2) - 5$

c. $g(x) = f(x + 2) + 5$

d. $g(x) = f(x - 2) + 5$

31 [Intermediate] • Target: Domain and range

The graph of function g is obtained by shifting the graph of f 4 units to the right and 7 units up. Which of the following defines $g(x)$?

a. $g(x) = f(x + 4) - 7$

b. $g(x) = f(x - 4) - 7$

c. $g(x) = f(x - 4) + 7$

d. $g(x) = f(x + 4) + 7$

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32 [Intermediate] • Target: Composition of functions

The graph of function g is obtained by shifting the graph of f 4 units to the left and 7 units up. Which of the following defines $g(x)$?

a. $g(x) = f(x + 4) - 7$

b. $g(x) = f(x - 4) + 7$

c. $g(x) = f(x - 4) - 7$

d. $g(x) = f(x + 4) + 7$

33 [Intermediate] • Target: Inverse functions

What is the maximum value of the function $f(x) = -x^2 + 10x - 20$?

a. 9

b. -5

c. 1

d. 5

34 [Intermediate] • Target: Transformations of graphs

What is the minimum value of the function $f(x) = 2x^2 - 20x + 53$?

a. 5

b. 7

c. 3

d. -5

35 [Intermediate] • Target: Piecewise functions

The graph of $f(x)$ is shifted 2 units left and 3 units up. Which expression gives the new function?

a. $f(x - 2) + 3$

b. $f(x + 2) + 3$

c. $f(x - 2) - 3$

d. $f(x + 2) - 3$

36 [Intermediate] • Target: Reading function graphs

What value of x is excluded from the domain of $f(x) = \frac{1}{x+3}$?

a. 3

b. -3

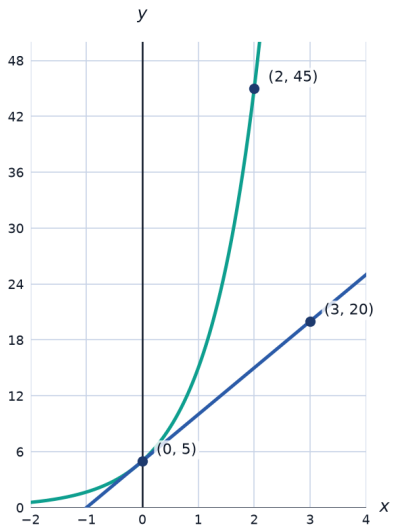
c. -2

d. -1

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37 **[Intermediate]** • **Target: Even, odd, and other function properties**

The diagram shows the graphs of the exponential function $y = a \cdot b^x$ and the linear function $y = mx + c$. The exponential function passes through the points $(0, 5)$ and $(2, 45)$. The linear function passes through the points $(0, 5)$ and $(3, 20)$. Find the value of a.



a. 1

b. 5

c. 3

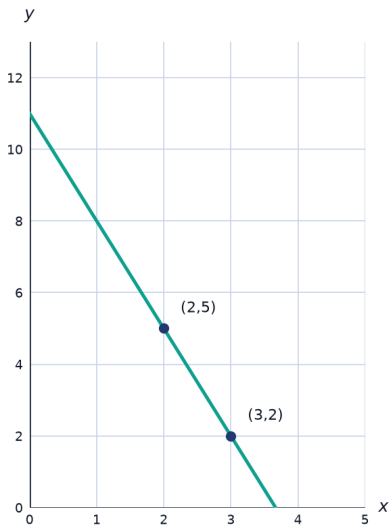
d. 9

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: A well-designed word problem: modeling a car rental cost** • **Cross-Lesson: 7.01 Trigonometry & Additional Topics**

In the xy-plane, the perpendicular lines T and K intersect at the point (3,2). If line T contains the point (2,5), which of the following points is on line K?



a. (5,3)

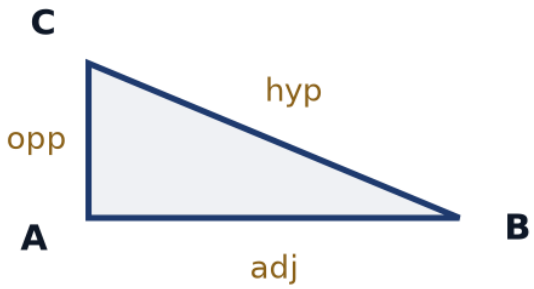
b. (-2,1)

c. (1,2)

d. (6,3)

39 **[Challenging]** • **Target: Evaluating functions** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

What is the y-intercept of the line passing through points (5;5) and (-5;-1)?



a. -3

b. 2

c. 0

d. 4

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40 [Challenging] • Target: Domain and range • Cross-Lesson: 1.01 Heart of Algebra

Consider the functions $f(x) = 2x + 3$ and $g(x) = x + 7$. At what x-value do the graphs of $f(x)$ and $g(x)$ intersect?

a. 4

b. 7

c. 3

d. 10

41 [Challenging] • Target: Composition of functions • Cross-Lesson: 2.01 Quadratics & Polynomials

If $f(x) = \frac{6x - 4}{2}$, which of the following represents $f^{-1}(x)$?

a. $\frac{2x + 4}{6}$

b. $\frac{6}{2x - 4}$

c. $\frac{2x - 4}{6}$

d. $\frac{6x + 4}{2}$

42 [Challenging] • Target: Inverse functions • Cross-Lesson: 4.01 Ratios, Rates & Data

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = 4x^2 + 4$. What is the value of $f(g(4))$?

a. 104

b. 137

c. 133

d. 123

43 [Challenging] • Target: Transformations of graphs • Cross-Lesson: 5.01 Statistics & Probability

At what x-value do $f(x) = 1x + 8$ and $g(x) = 3x - 2$ intersect?

a. -5

b. 5

c. 13

d. 6

44 [Challenging] • Target: Piecewise functions • Cross-Lesson: 6.01 Geometry

The graph of $y = (x - 9)^2$ is the graph of $y = x^2$ shifted how?

a. 9 units up

b. 9 units left

c. 10 units right

d. 9 units right

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45 [Challenging] • Target: Reading function graphs • Cross-Lesson: 7.01 Trigonometry & Additional Topics

The table shows a company's annual revenue (in 1000s) for two years. Assuming the relationship is linear, what is the rate of change in revenue per year between 2013 and 2019?

a. 15,000/yr

b. 16,000/yr

c. 18,000/yr

d. 20,000/yr

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